Exercise J - A

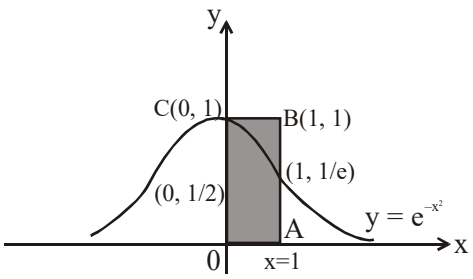
1. Let S be the area of the region enclosed by $y = e^{-x^2}, y = 0, x = 0$ and $x = 1$. Then : **[JEE-2012]**

यदि S वक्रों $y = e^{-x^2}, y = 0, x = 0$ तथा $x = 1$ के मध्य घिरा हुआ क्षेत्रफल हो तो : **[JEE-2012]**

(A) $S \geq \frac{1}{e}$ (B) $S \geq 1 - \frac{1}{e}$ (C) $S \leq \frac{1}{4}\left(1 + \frac{1}{\sqrt{e}}\right)$ (D) $S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}}\left(1 - \frac{1}{\sqrt{2}}\right)$

Ans. ABD

Sol. Area (OABC) = 1



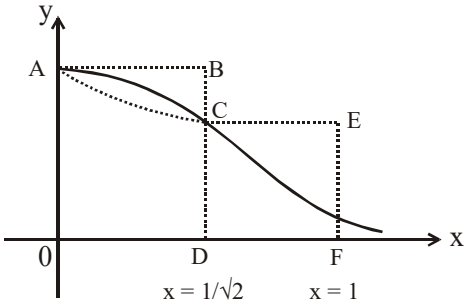
Shaded area is S.

Clearly S < 1

and $\int_0^1 e^{-x^2} dx > \int_0^1 e^{-x} dx$

$\Rightarrow \quad S > 1 - \frac{1}{e}$ ($\therefore$ (B) is correct

Again S ≥ Area (trapezium ACDO)



$\Rightarrow \quad S \geq \frac{1}{2}\left(1 + \frac{1}{\sqrt{e}}\right)\left(\frac{1}{\sqrt{2}}\right)$

$\Rightarrow \quad S \geq \frac{1}{2\sqrt{2}}\left(1 + \frac{1}{\sqrt{e}}\right)$

$\therefore \quad$ C is wrong

Also S ≤ Sum of areas of rectangles ABDO & CEFD

$\Rightarrow \quad S \leq \frac{1}{\sqrt{2}} \times 1 + \left(1 - \frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{e}}\right)$

$\Rightarrow \quad S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}}\left(1 - \frac{1}{\sqrt{2}}\right)$

($\therefore$ (D) is correct)

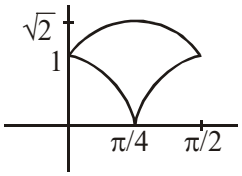
2. The area enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ over the interval $\left[0, \frac{\pi}{2}\right]$ is : **[JEE(Adv.)2013]**

अन्तराल $\left[0,\frac{\pi}{2}\right]$ पर वक्रों $y = \sin x + \cos x$ तथा $y = |\cos x - \sin x|$ द्वारा परिबद्ध क्षेत्रफल है : **[JEE(Adv.)2013]**

(A) $4\left(\sqrt{2}-1\right)$ (B) $2\sqrt{2}\left(\sqrt{2}-1\right)$ (C) $2\left(\sqrt{2}+1\right)$ (D) $2\sqrt{2}\left(\sqrt{2}+1\right)$

Ans. B

Sol. $y = \sin x + \cos x = \sqrt{2}\sin\left(x+\frac{\pi}{4}\right)$



$$y = |\cos x - \sin x| = \sqrt{2}\left(\cos\left(x+\frac{\pi}{4}\right)\right)$$

Area

$$= \int_0^{\pi/4}\left[\left(\sin x+\cos x\right)-\left(\cos x-\sin x\right)\right]dx$$

$$+\int_{\pi/4}^{\pi/2}\left[\left(\sin x+\cos x\right)-\left(\sin x-\cos x\right)\right]dx$$

$$= \int_0^{\pi/4} 2\sin x\,dx + \int_{\pi/4}^{\pi/2} 2\cos x\,dx$$

$$= \left[-2\cos x\right]_0^{\pi/4} + \left[2\sin x\right]_{\pi/4}^{\pi/2}$$

$$= 2\sqrt{2}\left(\sqrt{2}-1\right)$$

3. Let $f:\left[\frac{1}{2},1\right]\to R$ (the set of all real numbers) be a positive, non-constant and differentiable function such that $f'(x) < 2f(x)$ and $f\left(\frac{1}{2}\right)=1$. Then the value of $\int_{1/2}^{1} f(x)\,dx$ lies in the interval : **[JEE(Adv.)2013]**

माना कि $f:\left[\frac{1}{2},1\right]\to R$ (सभी वास्तविक संख्याओं का समुच्चय) एक धनात्मक अचरेतर (non constant) तथा अवकलनीय फलन gSft 1 ds fy ; sf'(x) < 2 f(x) तथा $f\left(\frac{1}{2}\right)=1$ है, तब $\int_{1/2}^{1} f(x)\,dx$ का मान निम्न अन्तराल में है : **[JEE(Adv.)2013]**

(A) $(2e-1, 2e)$ (B) $(e-1, 2e-1)$ (C) $\left(\frac{e-1}{2}, e-1\right)$ (D) $\left(0, \frac{e-1}{2}\right)$

Ans. D

Sol. $f'(x) - 2f(x) < 0$

Multiply both side by e^{-2x}

$$e^{-2x} f'(x) - 2e^{-2x} f(x) < 0$$

$$\frac{d}{dx}\left(e^{-2x} f(x)\right) < 0$$

Now, $g(x) = e^{-2x} f(x)$

$\therefore$ g(x) is a decreasing function.

$$x > \frac{1}{2}$$

$$g(x) < g\left(\frac{1}{2}\right)$$

$$\Rightarrow \quad e^{-2x} f(x) < \frac{1}{e}$$

$$\Rightarrow \quad f(x) < e^{2x-1}$$

$$\Rightarrow \quad \int_{1/2}^{1} f(x)dx < \frac{1}{e}\int_{1/2}^{1} e^{2x}dx$$

$$= \left[\frac{1}{2e}e^{2x}\right]_{1/2}^{1} = \frac{1}{2e}\left(e^2 - e\right) = \frac{1}{2}\left(e-1\right)$$

$$\Rightarrow \quad \int_{1/2}^{1} f(x)dx < \frac{e-1}{2}$$

obviously $f(x)$ is positive

$$\therefore \quad \int_{1/2}^{1} f(x)dx > 0$$

4. Let $F(x) = \int_{x}^{x^2+\frac{\pi}{6}} 2\cos^2 t\, dt$ for all $x \in R$ and $f: \left[0, \frac{1}{2}\right] \to [0, \infty)$ be a continuous function. For $a \in \left[0, \frac{1}{2}\right]$, if $F'(a) + 2$ is the area of the region bounded by $x = 0$, $y = 0$, $y = f(x)$ and $x = a$, then $f(0)$ is. **[JEE(Adv.)2015]**

माना कि सभी $x \in R$ के लिए, $F(x) = \int_{x}^{x^2+\frac{\pi}{6}} 2\cos^2 t\,dt$ तथा $f: \left[0, \frac{1}{2}\right] \to [0, \infty)$ एक संतत फलन है। यदि उन सभी $a \in \left[0, \frac{1}{2}\right]$ के लिए $F'(a) + 2$ उस क्षेत्र का क्षेत्रफल है, जो कि $x = 0$, $y = 0$, $y = f(x)$ और $x = a$ से घिरा हुआ है, तब $f(0)$ का मान है।

[JEE(Adv.)2015]

Ans. 3

Sol. $F(x) = \int_{x}^{x^2+\frac{\pi}{6}} 2\cos^2 + dt$

$$F'(x) = 2\left(\cos\left(x^2 + \frac{\pi}{6}\right)\right)^2 2x - 2\cos^2 x$$

$$\because F'(a) + 2 = \int_0^a f(x)dx$$

$$2\left(\cos\left(a^2 + \frac{\pi}{6}\right)\right)^2 2a - 2\cos^2 a + 2 = \int_0^a f(x)dx$$

$$\Rightarrow 4\cos^2\left(a^2 + \frac{\pi}{6}\right) + 4a2\cos\left(a^2 + \frac{\pi}{6}\right).\left(-\sin\left(a^2 + \frac{\pi}{6}\right)\right) \times 2a + 4\cos a \sin a = f(a)$$

$$\therefore f(0) = 4\left(\frac{\sqrt{3}}{2}\right)^2 = 3$$

5. Match the following column : **[JEE(Adv.)-2015]**

Column I	**Column II**
(A) In a triangle ΔXYZ, let a, b and c be the lengths of the sides opposite to the angles X, Y and Z, respectively. If $2(a^2 - b^2) = c^2$ and $\lambda = \frac{\sin(X-Y)}{\sin Z}$, then possible values of n for which $\cos(n\pi\lambda) = 0$ is (are)	(P) 1
(B) In a triangle ΔXYZ, let a, b and c be the lengths of the sides opposite to the angles X, Y and Z, respectively. If $1 + \cos 2X - 2\cos 2Y = 2 \sin X \sin Y$, then possible value(s) of a/b is (are)	(Q) 2
(C) In R^2, let $\sqrt{3}\,\hat{i} + \hat{j}, \hat{i} + \sqrt{3}\,\hat{j}$ and $\beta\,\hat{i} + (1-\beta)\hat{j}$ be the position vectors of X, Y and Z with respect to the origin O, respectively. If the distance of Z	(R) 3

from the bisector of the acute angle of $\overrightarrow{OX}$ with $\overrightarrow{OY}$ is $\frac{3}{\sqrt{2}}$, then possible value(s) of $|\beta|$ is (are)

(D) Suppose that $F(\alpha)$ denotes the area of the region bounded by $x = 0$, $x = 2$, $y^2 = 4x$ and $y = |\alpha x - 1| + |\alpha x - 2| + \alpha x$, where $\alpha \in \{0, 1\}$. Then the value(s) of $F(\alpha) + \frac{8}{3}\sqrt{2}$, when $\alpha = 0$ and $\alpha = 1$, is (are) (S) 5

(T) 6

निम्न कॉलम कों सुमेहिलत कीजिए : **[JEE(Adv.)-2015]**

Column I	**Column II**
(A) माना कि एक त्रिभुज ΔXYZ में कोणों X, Y और Z के सामने की भुजाओं की लम्बाईयाँ क्रमशः a, b और c है। मानाकि $2(a^2 - b^2) = c^2$ और $\lambda = \frac{\sin(X-Y)}{\sin Z}$ है। यदि $\cos(n\pi\lambda) = 0$ तब n के संभव मान है।	(P) 1
(B) माना कि एक त्रिभुज ΔXYZ मे कोणों X, Y और Z के सामने की भुजाओं की लम्बाईयाँ क्रमशः a, b और c है। यदि $1 + \cos 2X - 2\cos 2Y = 2 \sin X \sin Y$ तब a/b के संभव मान है	(Q) 2
(C) माना कि R^2 में, मूल बिन्दु O के सापेक्ष $\sqrt{3}\,\hat{i} + \hat{j}, \hat{i} + \sqrt{3}\,\hat{j}$ और $\beta\,\hat{i} + (1-\beta)\hat{j}$ क्रमशः र एल और Z के स्थिति सदिश है। यदि $\overrightarrow{OX}$ और $\overrightarrow{OY}$ के न्यून कोण के समद्विभाजक से Z की दूरी $\frac{3}{\sqrt{2}}$ हो, तो $\|\beta\|$ का संभव मान है।	(R) 3
(D) माना कि $F(\alpha)$ उस क्षेत्र के क्षेत्रफल को दर्शाता है जो $x = 0$, $x = 2$, $y^2 = 4x$ और $\psi = \|\alpha x - 1\| + \|\alpha x - 2\| + \alpha x$, से घिरा हुआ है, जहाँ $\alpha \in \{0, 1\}$ है। $\alpha = 0$ और $\alpha = 1$ के लिए $F(\alpha) + \frac{8}{3}\sqrt{2}$ का मान है।	(S) 5
	(T) 6

Ans. A -PRS , B -P , C - PQ, D - ST

Sol. (A) $2(a^2 - b^2) = c^2$

$2(4R^2(\sin^2 x \times - \sin^2 y)) = 4R^2 \sin^2 z$

$2(\sin(x-y)\sin(x+y)) = \sin^2 z$

$\frac{\sin(x-y)}{\sin z} = \frac{1}{2} = \lambda$

$$\cos(n\pi\lambda) = \cos\left(\frac{n\pi}{2}\right) = 0$$

n = odd integer

(B) $1 + \cos 2x = 2\cos 2y = 2\sin x \sin y$

$$\sin^2 x + \sin x \sin y - 2\sin^2 y = 0$$

$$(\sin x - \sin y)(\sin x + 2\sin y) = 0$$

$\sin x = \sin y$

a = b

$$\frac{a}{b} = 1$$

(C) distance of z from bisector of $\overrightarrow{OX}$ & $\overrightarrow{OY}$ = $\frac{3}{\sqrt{2}}$

$$\left(\beta - \frac{1}{2}\right)^2 + \left(\beta - \frac{1}{2}\right)^2 = \frac{9}{2}$$

$\beta = 2, -1$

$|\beta| = 2, 1$

(D) when $\alpha = 0$

$$A = 6 - \int_0^2 2\sqrt{x}\,dx$$

$$= 6 - \frac{8\sqrt{2}}{3}$$

when $\alpha = 1$

$$A = \int_0^1 \left(3 - x - 2\sqrt{x}\right)dx + \int_1^2 \left(x + 1 - 2\sqrt{x}\right)dx$$

$$= \left|3x - \frac{x^2}{2} - \frac{4}{3}x^{\frac{3}{2}}\right|_0^1 + \left|\frac{x^2}{2} + x - \frac{4}{3}x^{\frac{3}{2}}\right|_1^2$$

$$= 5 - \frac{8\sqrt{2}}{3}$$

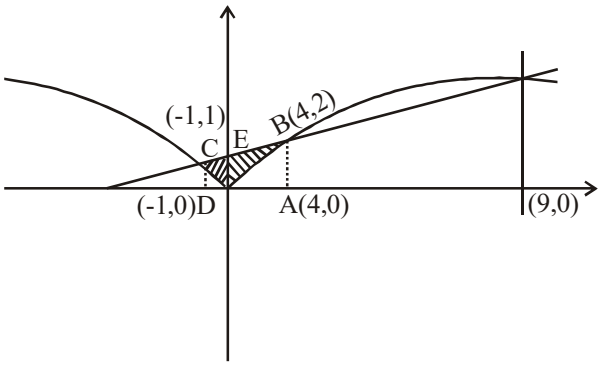
6. Area of the region $\left\{(x, y) \in R^2 : y \geq \sqrt{|x+3|}, 5y \leq x+9 \leq 15\right\}$ is equal to : **[JEE(Adv.)2016]**

क्षेत्र $\left\{(x, y) \in R^2 : y \geq \sqrt{|x+3|}, 5y \leq x+9 \leq 15\right\}$ का क्षेत्रफल है : **[JEE(Adv.)2016]**

(A) $\frac{1}{6}$ (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) $\frac{5}{3}$

Ans. C

Sol. Shifting origin to (–3, 0)



Area $\left\{(x, y) \in R^2 : y \geq \sqrt{|x|}, 5y \leq x+6 \leq 15\right\}$

Desired Area = Area of trapezium ABCDA – [Area DCOD + Area OBAO]

$$= \frac{1}{2} \times (1+2) \times 5 - \left(\int_{-1}^{0} \sqrt{-x}\ dx + \int_{0}^{4} \sqrt{x}\ dx \right)$$

$$= \frac{15}{2} - \left(\frac{2}{3} + \frac{16}{3} \right)$$

$$= \frac{3}{2}$$

7. If the line $x = \alpha$ divides the area of region $R = \{(x, y) \in R^2 : x^3 \leq y \leq x, 0 \leq x \leq 1\}$ into two equal parts, then : **[JEE(Adv.)-2017]**

यदि रेखा $x = \alpha$ क्षेत्र $R = \{(x, y) \in R^2 : x^3 \leq y \leq x, 0 \leq x \leq 1\}$ के क्षेत्रफल को दो बराबर भागों में विभाजित करती है, तब : **[JEE(Adv.)-2017]**

(A) $0 < \alpha \le \frac{1}{2}$ (B) $2\alpha^4 - 4\alpha^2 + 1 = 0$ (C) $\alpha^4 + 4\alpha^2 - 1 = 0$ (D) $\frac{1}{2} < \alpha < 1$

Ans. BD

Sol. $y = x^3$

$$\int_0^1 \left(x - x^3\right)dx = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$\int_0^\alpha \left(x - x^3\right)dx = \frac{1}{8}$$

$$4\alpha^2 - 2\alpha^4 = 1$$

$$2\alpha^4 - 4\alpha^2 + 1 = 0$$

$2t^2 - 4t + 1 = 0$ (taking $t = \alpha^2$)

$$t = \frac{4 \pm \sqrt{16-8}}{4}$$

$$t = \frac{4 \pm 2\sqrt{2}}{4}$$

$$t = \alpha^2 = 1 \pm \frac{1}{\sqrt{2}}$$

$$\therefore \alpha^2 = 1 - \frac{1}{\sqrt{2}} \qquad \Rightarrow \frac{1}{2} < \alpha < 1$$

8. Let $f: [0, \infty) \to R$ be a continuous function such that $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)\,dt$ for all $x \in [0, \infty)$. Then, which of the following statement(s) is (are) TRUE ? **[JEE(Adv.)2018]**

(A) The curve $y = f(x)$ passes through the point $(1, 2)$

(B) The curve $y = f(x)$ passes through the point $(2, -1)$

(C) The area of the region $\left\{(x, y) \in [0,1] \times R : f(x) \le y \le \sqrt{1-x^2}\right\}$ is $\frac{\pi - 2}{4}$

(D) The area of the region $\left\{(x, y) \in [0,1] \times R : f(x) \le y \le \sqrt{1-x^2}\right\}$ is $\frac{\pi - 1}{4}$

माना कि $f: [0, \infty) \to R$ एक ऐसा संतत फलन है कि $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)\,dt$ सभी $x \in [0, \infty)$ के लिये। तब निम्नलिखित esa lsa dkSu l k(से) कथन सत्य है (हैं)? **[JEE(Adv.)2018]**

(A) वक्र $y = f(x)$ बिन्दु (1, 2) से गुजरता है

(B) वक्र $y = f(x)$ बिन्दु (2, –1) से गुजरता है

(C) क्षेत्र $\left\{(x,y) \in [0,1] \times R : f(x) \le y \le \sqrt{1-x^2}\right\}$ का क्षेत्रफल $\frac{\pi-2}{4}$ है

(D) क्षेत्र $\left\{(x,y) \in [0,1] \times R : f(x) \le y \le \sqrt{1-x^2}\right\}$ का क्षेत्रफल $\frac{\pi-1}{4}$ है

Ans. BC

Sol. $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)dt \Rightarrow f(x).e^{-x} = (1-2x).e^{x} + \int_0^x e^{-t} f(t)dt$

$\Rightarrow f'(x)e^{-x} - e^{-x} \cdot f(x) = -2 \cdot e^{-x} - (1-2x) \cdot e^{-x} + e^{-x} \cdot f(x)$

$\Rightarrow f'(x) - 2f(x) = (2x-3)$

I.F. $= e^{-2x}$

$\therefore y.e^{-x} = \int \underset{|}{(2x-3)}.\underset{||}{e^{-2x}}dx$

$\Rightarrow y.e^{-2x} = (2x-3).\frac{e^{-2x}}{-2} - 2\int \frac{e^{-2x}}{-2}dx \qquad \Rightarrow y.e^{-2x} = -\frac{(2x-3)e^{-2x}}{2} - \frac{e^{-2x}}{2} + c$

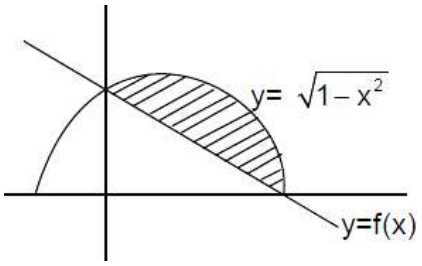
$\Rightarrow y.e^{-2x} = -\frac{(2x-3)-1}{2} + c.e^{2x} \qquad \Rightarrow y = (1-x) + c.e^{2x}$

$\Rightarrow y = (1-x) + c.e^{2x}$

put $x = 0$

$1 = 1 + c \qquad \Rightarrow c = 0$

$\therefore y = 1 - x$ which passes through point (2, –1)



Now

required area $= \frac{1}{4}.\pi.(1)^2 - \frac{1}{2}.1.1 = \frac{\pi}{4} - \frac{1}{2}$

9. A farmer F_1 has a land in the shape of a triangle with vertices at P(0,0), Q(1,1) and R(2,0). From this land, a neighbouring farmer F_2 takes away the region which lies between the side PQ and a curve of the form $y = x^n$ $(n > 1)$. If the area of the region taken away by the farmer F_2 is exactly 30% of the area of ΔPQR, then the value of n is _____ . **[JEE(Adv.)2018]**

एक किसान F_1 के पास एक त्रिभुजाकार भूमि है जिसके शीर्ष P(0,0), Q(1,1) तथा R(2,0) पर हैं। एक पड़ोसी किसान F_2 इस भूमि से उस क्षेत्र को ले लेता है जो कि भुजा PQ और $y = x^n$ $(n > 1)$ के रूप वाले वक्र के बीच स्थित है। यदि किसान F_2 द्वारा लिये गये क्षेत्र का क्षेत्रफल ΔPQR के क्षेत्रफल का ठीक 30% है, तब n का मान है। **[JEE(Adv.)2018]**

Ans. 4

Sol. $\frac{30}{100}A = \frac{3}{10}$

$$A = \int_0^1 \left(x - x^n\right) dx = \frac{3}{10}$$

$$\frac{x^2}{2} - \frac{x^{n+1}}{n+1}\Bigg|_0^1 = \frac{3}{10}$$

$$\frac{1}{2} - \frac{1}{n+1} = \frac{3}{10}$$

$n = 4$

10. The area of the region $\{(x,y): xy \le 8, 1 \le y \le x^2\}$ is : **[JEE(Adv.)2019]**

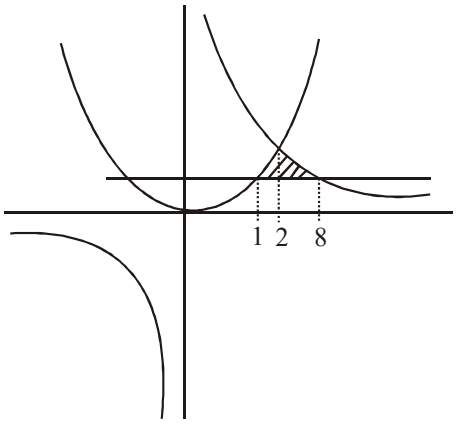
क्षेत्र $\{(x,y): xy \le 8, 1 \le y \le x^2\}$ का क्षेत्रफल है : **[JEE(Adv.)2019]**

(A) $16\log_e 2 - 6$ (B) $8\log_e 2 - \frac{14}{3}$ (C) $16\log_e 2 - \frac{14}{3}$ (D) $8\log_e 2 - \frac{7}{3}$

Ans. C

Sol. $x^3 = 8$

$x = 2$



$$\text{Area} = \int_1^2 \left(x^2 - 1\right) dx + \int_2^8 \left(\frac{8}{x} - 1\right) dx$$

$$= \left(\frac{x^3}{3} - x\right)_1^2 + \left(8\ln x - x\right)_2^8$$

$$= \left(\frac{8}{3} - 2 - \frac{1}{3} + 1\right) + \left[8\ln 8 - 8\ln 2 - 8 + 2\right]$$

$$= \left(\frac{7}{3} - 1\right) + 8\ln 4 - 6$$

$$= 16\ln 2 + \frac{7}{3} - 7$$

$$= 16\ln 2 - \frac{14}{3}$$

11. Let the functions $f: \mathbb{R} \to \mathbb{R}$ and $g: \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = e^{x-1} - e^{-|x-1|}$ and $g(x) = \frac{1}{2}(e^{x-1} + e^{1-x})$. Then the area of the region in the first quadrant bounded by the curves $y = f(x)$, $y = g(x)$ and $x = 0$ is :

[JEE(Adv.)2020]

माना कि फलनों $f: \mathbb{R} \to \mathbb{R}$ एवं $g: \mathbb{R} \to \mathbb{R}$ को $f(x) = e^{x-1} - e^{-|x-1|}$ एवं $g(x) = \frac{1}{2}(e^{x-1} + e^{1-x})$ के द्वारा परिभाषित किया जाता

gSr c i Ze pr dkZkeksa Øks $y = f(x)$, $y = g(x)$ एवं $x = 0$ के द्वारा प्रतिबद्ध क्षेत्र का क्षेत्रफल है? **[JEE(Adv.)2020]**

(A) $\left(2-\sqrt{3}\right)+\frac{1}{2}\left(e-e^{-1}\right)$

(B) $\left(2+\sqrt{3}\right)+\frac{1}{2}\left(e-e^{-1}\right)$

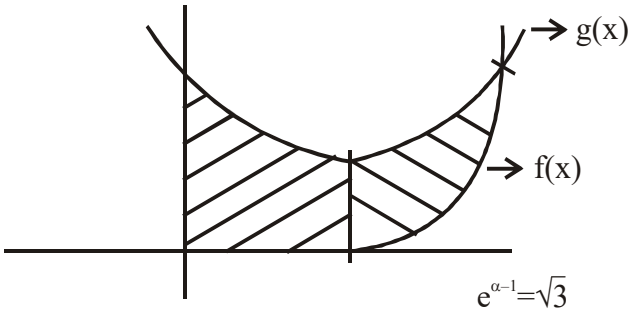
(C) $\left(2-\sqrt{3}\right)+\frac{1}{2}\left(e+e^{-1}\right)$

(D) $\left(2+\sqrt{3}\right)+\frac{1}{2}\left(e+e^{-1}\right)$

Ans. A

Sol. $f(x) \begin{cases} \to e^{x-1} - e^{1-x} & x \geq 1 \\ \to 0 & x < 1 \end{cases}$

$g(x) = \frac{1}{2}\left(e^{x-1} + e^{1-x}\right)$



$P\left(1+ln\sqrt{3}, \frac{2}{\sqrt{3}}\right) \equiv P\left(\alpha, \frac{2}{\sqrt{3}}\right)$

$\text{Required Area} = \int_0^1 \frac{e^{x-1}+e^{1-x}}{2}\,dx + \int_1^\alpha f(x) - f(x)\,dx$

$= \frac{1}{2}\left(e-\frac{1}{e}\right) + \int_1^\alpha \left(\frac{e^{x-1}+e^{1-x}}{2} - \left(e^{x-1}-e^{1-x}\right)\right)dx$

$= \frac{1}{2}\left(e-\frac{1}{e}\right) + \int_1^\alpha \left(\frac{3}{2}e^{1-x} - \frac{e^{x-1}}{2}\right)dx = \left(2-\sqrt{3}\right)+\frac{1}{2}\left(e-e^{-1}\right)$

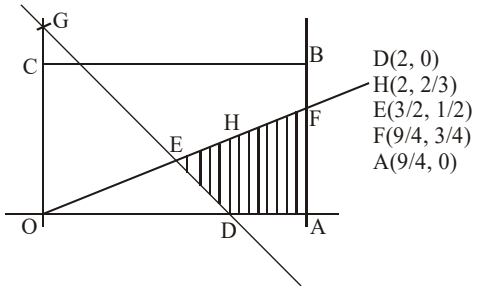
12. The area of the region $\left\{(x,y): 0 \leq x \leq \frac{9}{4},\quad 0 \leq y \leq 1,\quad x \geq 3y,\quad x+y \geq 2\right\}$ is : **[JEE(Adv.)2021]**

क्षेत्र $\left\{(x,y): 0 \leq x \leq \frac{9}{4},\quad 0 \leq y \leq 1,\quad x \geq 3y,\quad x+y \geq 2\right\}$ का क्षेत्रफल है : **[JEE(Adv.)2021]**

(A) $\frac{11}{32}$ (B) $\frac{35}{96}$ (C) $\frac{37}{96}$ (D) $\frac{13}{32}$

Ans. A

Sol.



$AB \equiv x = \frac{9}{4}$

$BC \equiv y = 1$

$DG \equiv x + y = 2$

$OF \equiv x = 3y$

$DG \equiv x = 2$

we will divide shaded region in two parts

required area = Area of ΔDEH + area of trapezium AFHD

$$= \frac{1}{2}\left(\frac{2}{3}\right)\left(\frac{1}{2}\right) + \frac{1}{2}\left(\frac{2}{3} + \frac{3}{4}\right)\left(\frac{1}{4}\right) = \frac{11}{32}$$

13. Consider the functions $f, g : R \to R$ defined by $f(x) = x^2 + \frac{5}{12}$ and $g(x) = \begin{cases} 2\left(1 - \frac{4|x|}{3}\right) & |x| \le \frac{3}{4}, \\ 0, & |x| > \frac{3}{4} \end{cases}$. If α is the area of the region $\left\{(x, y) \in R \times R : |x| \le \frac{3}{4}, 0 \le y \le \min\{f(x), g(x)\}\right\}$ then the value of 9α is ____. **[JEE(Adv.)2022]**

फलनों $f, g : R \to R$ पर विचार कीजिए जो $f(x) = x^2 + \frac{5}{12}$ एवं $g(x) = \begin{cases} 2\left(1 - \frac{4|x|}{3}\right) & |x| \le \frac{3}{4}, \\ 0, & |x| > \frac{3}{4} \end{cases}$ द्वारा परिभाषित है।

यदि क्षेत्र $\left\{(x, y) \in R \times R : |x| \le \frac{3}{4}, 0 \le y \le \min\{f(x), g(x)\}\right\}$ का क्षेत्रफल α है, तब 9α का मान _____ है।

[JEE(Adv.)2022]

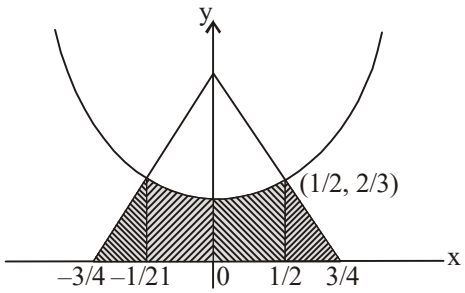
Ans. 6

Sol. $f(x)=x^2+\frac{5}{12}$

$g(x)=2\left(1-\frac{4}{3}|x|\right)$

$|x|\le\frac{3}{4}$ and $0\le y\le(f(x),g(x))$

for $x\ge 0$



$x^2+\frac{5}{12}=2-\frac{8}{3}x$

$\Rightarrow 12x^2+32x-19=0$

$12x^2+38x-6x-19=0\Rightarrow x=\frac{1}{2}$

Required area $=2\left(\int_0^{1/2}\left(x^2+\frac{5}{12}\right)dx+\frac{1}{2}\times\frac{1}{4}\times\left(2-\frac{4}{3}\right)\right)$

$\Rightarrow\alpha=2\left(\frac{1}{24}+\frac{5}{24}+\frac{2}{24}\right)=\frac{2}{3}$ sq.units $\Rightarrow 9\alpha=9\times\frac{2}{3}=6$

14. Let $f:[0,1]\to[0,1]$ be the function defined by $f(x)=\frac{x^3}{3}-x^2+\frac{5}{9}x+\frac{17}{36}$. Consider the square region $S=[0,1]\times[0,1]$. Let $G=\{(x,y)\in S:y>f(x)\}$ be called the green region and $R=\{(x,y)\in S:y<f(x)\}$ be called the red region. Let $L_h=\{(x,h)\in S:x\in[0,1]\}$ be the horizontal line drawn at a height $h\in[0,1]$. Then which of the following statements is(are) true? **[JEE(Adv.)2023]**

(A) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the green region above the line L_h equals the area of the green region below the line L_h

(B) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the red region above the line L_h equals the area of the red region below the line L_h

(C) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the green region above the line L_h equals the area of the red region below the line L_h

(D) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the red region above the line L_h equals the area of the green region below the line L_h

माना कि फलन $f:[0,1]\to[0,1]$, $f(x)=\frac{x^3}{3}-x^2+\frac{5}{9}x+\frac{17}{36}$ से परिभाषित है। वर्गाकार क्षेत्र(square region) $S=[0,1]\times[0,1]$ पर विचार कीजिए। माना कि $G=\{(x,y)\in S: y>f(x)\}$ हरित क्षेत्र(green region) एवं $R=\{(x,y)\in S: y<f(x)\}$ लाल क्षेत्र (red region) कहलाता है। मान लीजिये की $h\in[0,1]$ की ऊँचाई (height) पर खींची गई {kSfr t jsf[kk(horizontal line) $L_h=\{(x,h)\in S: x\in[0,1]\}$ है। तब निम्न में से कौनसा (से) कथन सत्य है(हैं)?

[JEE(Adv.)2023]

(A) एक ऐसा $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ है कि रेखा L_h के ऊपर के लाल क्षेत्र का क्षेत्रफल रेखा L_h के नीचे के हरित क्षेत्र के क्षेत्रफल के बराबर है

(B) एक ऐसा $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ है कि रेखा L_h के ऊपर के हरित क्षेत्र का क्षेत्रफल रेखा L_h के नीचे के लाल क्षेत्र के क्षेत्रफल के बराबर है

(C) एक ऐसा $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ है कि रेखा L_h के ऊपर के लाल क्षेत्र का क्षेत्रफल रेखा L_h के नीचे के लाल क्षेत्र के क्षेत्रफल के बराबर है

(D) एक ऐसा $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ है कि रेखा L_h के ऊपर के हरित क्षेत्र का क्षेत्रफल रेखा L_h के नीचे के हरित क्षेत्र के क्षेत्रफल के बराबर है

Ans. BCD

Sol. $f(x)=\frac{x^3}{3}-x^2+\frac{5x}{9}+\frac{17}{36}$

$f'(x)=x^2-2x+\frac{5}{9}$

$f'(x)=0$ at $x=\frac{1}{3}$ in $[0,1]$

$A_R=$ Area of Red region

$A_G=$ Area of Green region

$A_R=\int_0^1 f(x)\,dx=\frac{1}{2}$

Total area = 1

$\Rightarrow A_G=\frac{1}{2}$

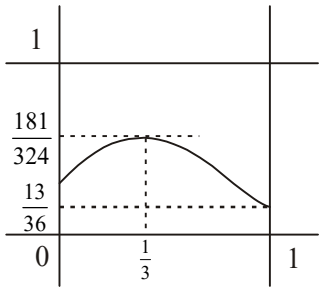
$\int_0^1 f(x)\,dx=\frac{1}{2}$

$A_G=A_R$

$f(0)=\frac{17}{36}$

$$f(1)=\frac{13}{36}\approx 0.36$$

$$f\left(\frac{1}{3}\right)=\frac{181}{324}\approx 0.558$$



(A) Given statement would be correct when $h=\frac{3}{4}$ but $h\in\left[\frac{1}{4},\frac{2}{3}\right]$

$\Rightarrow$ (A) is incorrect

(B) Given statement would be correct when $h=\frac{1}{4}$

$\Rightarrow$ (B) is correct

(C) When $h=\frac{181}{324}, A_R=\frac{1}{2}, A_G<\frac{1}{2}$

$h=\frac{13}{36}, A_R<\frac{1}{2}, A_G=\frac{1}{2}$

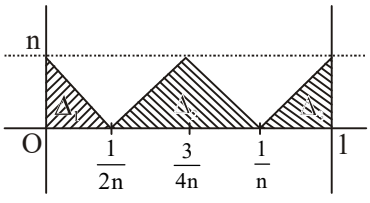
$\Rightarrow A_R=A_G$ for some $h\in\left(\frac{13}{36},\frac{181}{324}\right)$

$\Rightarrow$ (C) is correct

(D) Option (D) is remaining coloured part of option (C), hence option (D) is also correct.

15. Let $n\geq 2$ be a natural number and $f:[0,1]\to R$ be the function defined by

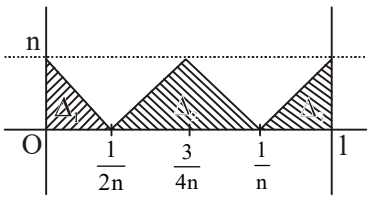
$$f(x)=\begin{cases} n(1-2nx) & \text{if } 0\leq x\leq\frac{1}{2n} \\ 2n(2nx-1) & \text{if } \frac{1}{2n}\leq x\leq\frac{3}{4n} \\ 4n(1-nx) & \text{if } \frac{3}{4n}\leq x\leq\frac{1}{n} \\ \frac{n}{n-1}(nx-1) & \text{if } \frac{1}{n}\leq x\leq 1 \end{cases}$$



If n is such that the area of the region bounded by the curves $x = 0$, $x = 1$, $y = 0$ and $y = f(x)$ is 4 , then the maximum value of the function f is _____. **[JEE(Adv.)2023]**

माना कि $n \geq 2$ एक प्राकृत संख्या (natural number) है एवं फलन $f : [0,1] \to R$ इस प्रकार परिभाषित है

$$f(x) = \begin{cases} n(1-2nx) & \text{if } 0 \leq x \leq \dfrac{1}{2n} \\ 2n(2nx-1) & \text{if } \dfrac{1}{2n} \leq x \leq \dfrac{3}{4n} \\ 4n(1-nx) & \text{if } \dfrac{3}{4n} \leq x \leq \dfrac{1}{n} \\ \dfrac{n}{n-1}(nx-1) & \text{if } \dfrac{1}{n} \leq x \leq 1 \end{cases}$$



यदि n इस प्रकार है कि वक्रों $x = 0$, $x = 1$, $y = 0$ एवं $y = f(x)$ से घिरे क्षेत्र का क्षेत्रफल 4 है तब फलन f का महत्तम मान (maximum value) है _____. **[JEE(Adv.)2023]**

Ans. 8

Sol. $$f(x) = \begin{cases} n(1-2nx) & 0 \leq n \leq \dfrac{1}{2n} \\ 2n(2nx-1) & \dfrac{1}{2n} \leq x \leq \dfrac{3}{4n} \\ 4n(1-nx) & \dfrac{3}{4n} \leq x \leq \dfrac{1}{n} \\ \dfrac{n}{n-1}(nx-1) & \dfrac{1}{n} \leq x \leq 1 \end{cases}$$

$\Delta_1 + \Delta_2 + \Delta_3 = 4$

$$\frac{1}{2} n \times \frac{1}{2n} + \frac{1}{2} \times n \times \frac{1}{2n} + \frac{1}{2} \times n \times \left(1 - \frac{1}{n}\right) = 4$$

$$4 = \frac{n}{2}$$

$n = 8$

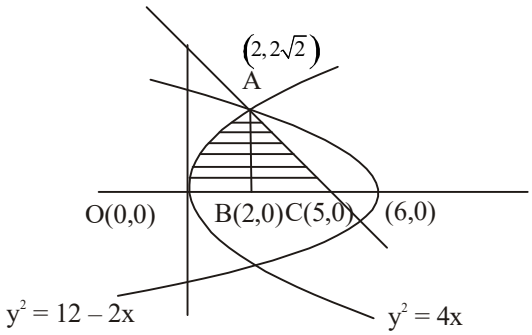
16. Let $S = \left\{(x, y) \in R \times R : x \geq 0,\ y \geq 0,\ y^2 \leq 4x,\ y^2 \leq 12 - 2x \text{ and } 3y + \sqrt{8}x \leq 5\sqrt{8}\right\}$ If the area of the region S is $\alpha\sqrt{2}$, then α is eqaul to : **[JEE(Adv.)2024]**

माना कि $S = \left\{(x, y) \in R \times R : x \geq 0,\ y \geq 0,\ y^2 \leq 4x,\ y^2 \leq 12 - 2x \text{ और } 3y + \sqrt{8}x \leq 5\sqrt{8}\right\}$ है। यदि क्षेत्र (region) S का {ksl Qy $\alpha\sqrt{2}$ है, तब α बराबर है : **[JEE(Adv.)2024]**

(A) $\frac{17}{2}$ (B) $\frac{17}{3}$ (C) $\frac{17}{4}$ (D) $\frac{17}{5}$

Ans. B

Sol. $3y+\sqrt{8}x=5\sqrt{8}$



$4x = 12 - 2x \Rightarrow 6x = 12 \Rightarrow x = 2$

$x = 2 \Rightarrow y^2 = 8 \Rightarrow y = \pm 2\sqrt{2}$

line $3y+\sqrt{8}x=5\sqrt{8}$ passes through $A\left(2,2\sqrt{2}\right)$ & C(5,0)

Required area = Area of OAB + Area of ΔABC

$$=\int_0^2 2\sqrt{x}dx+\frac{1}{2}(5-2)\left(2\sqrt{2}\right)$$

$$=\frac{4}{3}\left(x^{\frac{3}{2}}\right)_0^2+3\sqrt{2}$$

$$=\frac{8}{3}\sqrt{2}+3\sqrt{2}$$

$$=\frac{17}{3}\sqrt{2}$$

17. Let **R** denote the set of all real numbers. Then the area of the region

$$\left\{(x,y)\in \mathbf{R}\times\mathbf{R}: x>0, y>\frac{1}{x}, 5x-4y-1>0, 4x+4y-17<0\right\}$$

is: **[JEE(Adv.)2025]**

मान लीजिए कि **R** सभी वास्तविक संख्याओं (real numbers) के समुच्चय (set) को दर्शाता है। तब क्षेत्र (region)

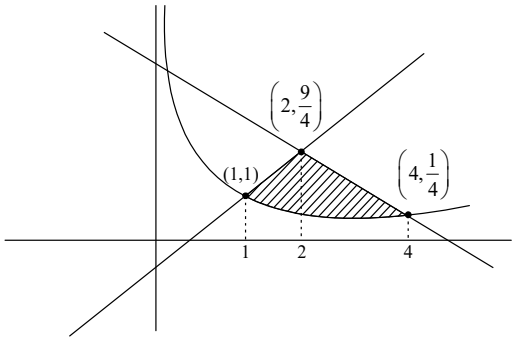
$$\left\{(x,y)\in \mathbf{R}\times\mathbf{R} : x>0, y>\frac{1}{x}, 5x-4y-1>0, 4x+4y-17<0\right\}$$

का क्षेत्रफल (area) है : **[JEE(Adv.)2025]**

(A) $\frac{17}{16}-\log_e 4$ (B) $\frac{33}{8}-\log_e 4$ (C) $\frac{57}{8}-\log_e 4$ (D) $\frac{17}{2}-\log_e 4$

Ans. B

Sol.



$$A=\frac{1}{2}\times\left(1+\frac{9}{4}\right)\times 1+\frac{1}{2}\times\left(\frac{9}{4}+\frac{1}{4}\right)\times 2-\int_1^4\frac{1}{x}dx$$

$$A=\frac{33}{8}-\ell n4$$

18. Let S denote the locus of the mid-points of those chords of the parabola $y^2 = x$, such that the area of the region enclossed between the parabola and the chord is $\frac{4}{3}$. Let R denote the region lying in the first quadrant, enclosed by the parabola $y^2 = x$, the curve S, and the lines $x = 1$ and $x = 4$.

Then which of the following statements is (are) TRUE? **[JEE(Adv.)2025]**

(A) $\left(4,\sqrt{3}\right)\in S$ (B) $\left(5,\sqrt{2}\right)\in S$

(C) Area of R is $\frac{14}{3}-2\sqrt{3}$ (D) Area of R is $\frac{14}{3}-\sqrt{3}$

मान लीजिए कि S, परवलय (parabola) $y^2 = x$ की उन सभी जीवाओं (chord) के मध्य–बिन्दुओं (mid-points) का बिन्दुपथ (locus) है, जिनके लिए परवलय एवं जीवा द्वारा घिरे क्षेत्र (region enclosed) का क्षेत्रफल (area) $\frac{4}{3}$ है। मान लीजिए कि R प्रथम चतुर्थांश (first quadrant) में उस क्षेत्र को दर्शाता है, जो परवलय $y^2 = x$, वक्र (curve) S, और रेखाओं $x = 1$ तथा $x = 4$ द्वारा घिरा हुआ है।

r c fuEufy f[kr d Hkqkesa sd kS l k¼ s1 R, gS½2 **[JEE(Adv.)2025]**

(A) $\left(4,\sqrt{3}\right)\in S$

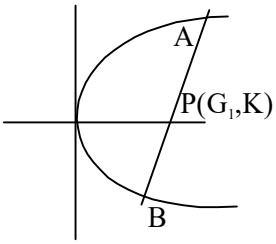
(B) $\left(5,\sqrt{2}\right)\in S$

(C) τ का क्षेत्रफल $\frac{14}{3}-2\sqrt{3}$ है

(D) R का क्षेत्रफल $\frac{14}{3}-\sqrt{3}$ है

Ans. AC

Sol. Equation of chord AB $T = S_1$



$$ky-\left(\frac{x+h}{2}\right)=k^2-h$$

$$x-2ky+2k^2-h=0$$

$$\Rightarrow h-k^2<0$$

For area interchange x & y

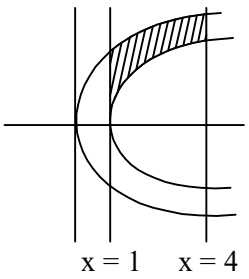
$$y-2kx+2k^2-h=0 \;\&\; y=x^2$$

$$x^2-2kx+\left(2k^2-h\right)=0\left\langle\begin{matrix}\alpha\\ \beta\end{matrix}\right.$$

$$|\alpha-\beta|=\sqrt{4k^2-4\left(2k^2-h\right)}=\sqrt{4h-4k^2}$$

$$A=\int_{\alpha}^{\beta}\left(\left(2kx+h-2k^2\right)-x^2\right)dx=\frac{4}{3}$$

$$\frac{\left(4h-4k^2\right)^{\frac{3}{2}}}{6}=\frac{4}{3}$$



$$\left(4h-4k^2\right)^{\frac{3}{2}}=8$$

$$\Rightarrow (4h-4k^2)=4$$

$$h-k^2=1$$

$$\left(4,\sqrt{3}\right)\in S$$

$$A=\int_{1}^{4}\left(\sqrt{x}-\sqrt{x-1}\right)dx=\frac{2}{3}\left[\left(x^{\frac{3}{2}}-\left(x-1\right)^{\frac{3}{2}}\right)\right]_1^4$$

$$=\frac{2}{3}\left(8-3\sqrt{3}-1\right)=\frac{14}{3}-2\sqrt{3}$$

Question Stem for Question Nos. 19 and 20

Consider the curve C_1 given by $y = e^{-x}$ for $x \in [0, 10\pi]$ and the curve C_2 given by $y = e^{-x}(\sin x + \cos x)$ for $x \in [0, 10\pi]$

Let n be the total number of points of intersection of the curves C_1 and C_2

Suppose that $\alpha_1, \alpha_2, \ldots\ldots\ldots, \alpha_n \in [0, 10\pi]$ are the x-coordinates of the points of intersection of the curves C_1 and C_2 such that $\alpha_1 < \alpha_2 < \ldots < \alpha_n$

वक्र C_1 जो $y = e^{-x}$ सभी $x \in [0, 10\pi]$ के लिए, द्वारा दिया गया है, और वक्र C_2 जो $y = e^{-x}(\sin x + \cos x)$ सभी $x \in [0, 10\pi]$ के लिए, द्वारा दिया गया है, पर विचार कीजिए।

मान लीजिए कि वक्रों C_1 और C_2 के कुल प्रतिच्छेद बिंदुओं की संख्या n है।

मान लीजिए कि $\alpha_1, \alpha_2, \ldots\ldots\ldots, \alpha_n \in [0, 10\pi]$, वक्रों C_1 और C_2 के प्रतिच्छेद बिंदुओं के x-निर्देशांक इस प्रकार हैं कि $\alpha_1 < \alpha_2 < \ldots < \alpha_n$ है।

19. The value of n is _____. **[JEE(Adv.)2026]**

n का मान _____ है। **[JEE(Adv.)2026]**

Ans. 11

Sol. To find the points of intersection $e^{-x} = e^{-x}(\sin x + \cos x)$

$1 = \sin x + \cos x$

$$\cos\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$x - \frac{\pi}{4} = 2k\pi + \frac{\pi}{4} \Rightarrow x = 2k\pi + \frac{\pi}{2} \text{ or } x - \frac{\pi}{4} = 2k\pi - \frac{\pi}{4} \Rightarrow x = 2k\pi \quad \text{where } k \in \mathbb{Z}$$

Now, we find the values of x that lie within the given domain $x \in [0, 10\pi]$

Φρομ $\xi = 2\kappa\pi$: $x = 0, 2\pi, 4\pi, 6\pi, 8\pi, 10\pi$ (6 points)

From $x = 2k\pi + \frac{\pi}{2}$: $x = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \frac{13\pi}{2}, \frac{17\pi}{2}$ (5 points)

$$\alpha_1 = 0, \alpha_2 = \frac{\pi}{2}, \alpha_3 = 2\pi, \alpha_4 = \frac{5\pi}{2}, \ldots, \alpha_{11} = 10\pi$$

Thus, the total number of intersection points is $n = 11$

20. Let β be the area of the region enclosed between the curves C_1, C_2 and the lines $x = \alpha_1$ and $x = \alpha_4$. Then the value

of $-\frac{1}{\pi}\log_e\left(\beta-2e^{-\frac{\pi}{2}}\right)$ is _____. **[JEE(Adv.)2026]**

मान लीजिए कि वक्रों C_1, C_2, और रेखाओं $x=\alpha_1$ और $x=\alpha_4$ से परिबद्ध क्षेत्र का क्षेत्रफल β है। तब $-\frac{1}{\pi}\log_e\left(\beta-2e^{-\frac{\pi}{2}}\right)$ का मान _____ है। **[JEE(Adv.)2026]**

Ans. 2.5

Sol. $\alpha_1=0$ and $\alpha_4=\frac{5\pi}{2}$

The enclosed area β is given by

$$\beta=\int_0^{\frac{5\pi}{2}}|y_{C_1}-y_{C_2}|dx=\int_0^{\frac{5\pi}{2}}e^{-x}\,|1-\sin x-\cos x|dx$$

First, let's determine the indefinite integral without absolute values

$$I=\int e^{-x}(\sin x+\cos x-1)dx=F(x)$$

$$F(x)=e^{-x}(1-\cos x)$$

Since the expression inside the absolute value changes sign only at the roots $(\alpha_1,\alpha_2,\alpha_3,\alpha_4)$, the total area β is the sum of the absolute differences of F(x) at consecutive boundaries

$$\beta=|F(\alpha_2)-F(\alpha_1)|+|F(\alpha_3)-F(\alpha_2)|+|F(\alpha_4)-F(\alpha_3)|$$

Substituting these values into the area formula

$$\beta=\left|e^{-\frac{\pi}{2}}-0\right|+\left|0-e^{-\frac{\pi}{2}}\right|+\left|e^{-\frac{5\pi}{2}}-0\right|$$

$$\beta=e^{-\frac{\pi}{2}}+e^{-\frac{\pi}{2}}+e^{-\frac{5\pi}{2}}=2e^{-\frac{\pi}{2}}+e^{-\frac{5\pi}{2}}$$

$$\beta-2e^{-\frac{\pi}{2}}=e^{-\frac{5\pi}{2}}$$

$$\log_e\left(\beta-2e^{-\frac{\pi}{2}}\right)=\log_e\left(e^{-\frac{5\pi}{2}}\right)=-\frac{5\pi}{2}$$

Question Stem for Question Nos. 21 and 22

Consider the ellipses given by : $x^2 + 4y^2 = 1$ and $4x^2 + y^2 = 1$

दीर्घवृत्तों : $x^2 + 4y^2 = 1$ और $4x^2 + y^2 = 1$ पर विचार कीजिए।

21. Let P be the point in the first quadrant where the given ellipses intersect. If θ is the acute angle between the tangents to the given ellipses at the point P, then the value of $4\tan\theta$ is _____. **[JEE(Adv.)2026]**

मान लीजिए कि P, प्रथम चतुर्थांश में वह बिन्दु है जहाँ दिए गये दीर्घवृत्त प्रतिच्छेद करते हैं। यदि θ, बिन्दु P पर दिए गये दीर्घवृत्तों की स्पर्श रेखाओं के बीच का न्यूनकोण है, तब $4\tan\theta$ का मान _____ होगा। **[JEE(Adv.)2026]**

Ans. 7.5

Sol. Given the equations of the two ellipses:

$x^2 + 4y^2 = 1$ and $4x^2 + y^2 = 1$

Subtracting equation (1) from equation (2):

$(4x^2 + y^2) - (x^2 + 4y^2) = 1 - 1$

$3x^2 - 3y^2 = 0 \quad \Rightarrow x^2 = y^2$

Since the point P lies in the first quadrant, both $x > 0$ and $y > 0$, so we have $x = y$.

$x^2 + 4x^2 = 1 \quad \Rightarrow 5x^2 = 1 \quad \Rightarrow x = \frac{1}{\sqrt{5}}$

Since $y = x$, the coordinates of point P are: $P = \left(\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$

Differentiating both equations with respect to x to find the slopes (m_1 and m_2):

For the first ellipse ($x^2 + 4y^2 = 1$): $2x + 8y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{4y}$

At $P\left(\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$, the slope m_1 is: $m_1 = -\frac{\frac{1}{\sqrt{5}}}{4\left(\frac{1}{\sqrt{5}}\right)} = -\frac{1}{4}$

For the second ellipse ($4x^2 + y^2 = 1$): $8x + 2y\frac{dy}{dx} = 0 \quad \Rightarrow \frac{dy}{dx} = -\frac{4x}{y}$

At $P\left(\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$, the slope m_2 is: $m_2 = -\frac{4\left(\frac{1}{\sqrt{5}}\right)}{\frac{1}{\sqrt{5}}} = -4$

The angle θ between two lines with slopes m_1 and m_2 is given by:

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| \Rightarrow \quad \tan\theta = \left|\frac{-\frac{1}{4} - (-4)}{1 + \left(-\frac{1}{4}\right)(-4)}\right| = \left|\frac{-\frac{1}{4} + 4}{1 + 1}\right| = \left|\frac{\frac{15}{4}}{2}\right| = \frac{15}{8}$$

$$4\tan\theta = 4 \times \frac{15}{8} = \frac{15}{2} = 7.5$$

22. If α is the area of the common region that lies inside both the given ellipses, then the value of $\cot\alpha$ is _____.

[JEE(Adv.)2026]

यदि α दिए गये दोनों दीर्घवृत्तों के अंदर के उभयनिष्ठ क्षेत्र का क्षेत्रफल है, तब $\cot\alpha$ का मान _____होगा।

[JEE(Adv.)2026]

Ans. 0.75

Sol. The given ellipses are highly symmetrical across all four quadrants and about the line $y = x$. We can convert their equations into polar coordinates ($x = r\cos\theta$, $y = r\sin\theta$) to easily integrate the bounded area.

First Ellipse : $r^2\cos^2\theta + 4r^2\sin^2\theta = 1 \qquad \Rightarrow r_1^2 = \frac{1}{1 + 3\sin^2\theta}$

Second Ellipse: $4r^2\cos^2\theta + r^2\sin^2\theta = 1 \qquad \Rightarrow r_2^2 = \frac{1}{4 - 3\sin^2\theta}$

In the first quadrant, the two curves intersect at $\theta = \frac{\pi}{4}$. Due to symmetry, the total area α is 8 times the area bounded from $\theta = 0$ to $\theta = \frac{\pi}{4}$ by the inner curve (r_2) :

$$\theta = 8 \times \left(\frac{1}{2}\int_0^{\pi/4} r_2^2\, d\theta\right) = 4\int_0^{\pi/4} \frac{1}{4 - 3\sin^2\theta} d\theta$$

$$\alpha = 4\int_0^{\pi/4} \frac{\sec^2\theta}{4\sec^2\theta - 3\tan^2\theta} d\theta \Rightarrow \alpha = 4\int_0^{\pi/4} \frac{\sec^2\theta}{4(1 + \tan^2\theta) - 3\tan^2\theta} d\theta = 4\int_0^{\pi/4} \frac{\sec^2\theta}{4 + \tan^2\theta} d\theta$$

Let $t = \tan\theta \qquad \Rightarrow dt = \sec^2\theta d\theta$

$$\alpha = 4\int_0^1 \frac{dt}{4 + t^2} = 4\left[\frac{1}{2}\tan^{-1}\left(\frac{t}{2}\right)\right]_0^1 \Rightarrow \quad \alpha = 2\tan^{-1}\left(\frac{1}{2}\right)$$

Let $\beta = \tan^{-1}\left(\frac{1}{2}\right)$. Thus, $\alpha = 2\beta$.

$$\tan\alpha = \tan(2\beta) = \frac{2\tan\beta}{1-\tan^2\beta} = \frac{2\left(\frac{1}{2}\right)}{1-\left(\frac{1}{2}\right)^2} = \frac{1}{1-\frac{1}{4}} = \frac{1}{\frac{3}{4}} = \frac{4}{3}$$

$$\cot\alpha = \frac{1}{\tan\alpha} = \frac{3}{4} = 0.75$$